PRODUCT REQUIREMENT DOCUMENT

(PRD)

DOMAIN 2

KeretaSewa AI: Decision Intelligence Platform for SME Car Rental Operators

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1. Project Overview

Problem Statement

In Malaysia, the car rental industry continues to grow alongside tourism and increasing demand for flexible transportation. The market is expected to exceed USD 1 billion in the coming years, reflecting a rising reliance on short term vehicle access by both locals and travelers (Mordor Intelligence, 2024).

However, despite this growth, many small and medium sized car rental businesses still rely on basic tools such as spreadsheets, WhatsApp, or simple booking systems. While these tools are effective for recording bookings and managing communication, they lack analytical capabilities such as tracking trends, identifying patterns, or generating insights.

As a result, several key challenges arise:

- **Low Fleet Utilization**
  Without proper analysis, operators cannot easily identify underperforming vehicles, resulting in cars remaining idle for long periods.
- **Inefficient Pricing Strategies**
  The absence of data insights forces operators to rely on intuition, leading to inconsistent or suboptimal pricing decisions.
- **Missed High Demand Opportunities**
  Without visibility into demand trends, operators may fail to adjust pricing or allocation during peak periods.
- **Reactive Decision Making**
  Decisions are made based on immediate situations rather than long term patterns, limiting overall optimization.

These challenges highlight that the core issue is not simply operational inefficiency, but a lack of decision intelligence , the ability to transform data into meaningful insights, actionable recommendations, and informed strategies.

Target Domain

Target Domain: Smart mobility and SME fleet management, focusing on car rental businesses that need AI driven insights for pricing, utilization, and operational decision making.

Proposed Solution Summary

KeretaSewa AI is an AI powered decision intelligence platform designed for small and medium sized car rental operators.

It acts as a decision making layer that analyzes operational data and provides:

- actionable recommendations
- clear explanations
- measurable business impact

2. Background & Business Objective

Background of the Problem

Existing digital tools used in the car rental industry often focus on reservation records, customer communication, scheduling, and basic transaction management. While these tools support day to day operations, they may not provide deeper analytical insights or decision support capabilities.

Although large organizations may use advanced analytics tools, these systems are typically designed for enterprise use. Small and medium sized enterprises (SMEs) often face limitations in terms of cost, technical complexity, and available resources, which restrict their ability to adopt such solutions (Omowole et al., 2024) .

In addition, fragmented operational data across multiple channels can reduce visibility and delay decision making. Studies show that effective use of analytics can improve efficiency, cost control, and resource utilization (McKinsey & Company, 2022).

Therefore, a clear gap exists between data availability and decision making capability. KeretaSewa AI addresses this gap by acting as a decision intelligence layer that transforms fragmented operational data into actionable recommendations, enabling SMEs to make informed and optimized decisions.

Importance of Solving This Issue

Addressing this problem can create meaningful and measurable business value for small and medium sized car rental operators. Many SMEs operate with limited resources, smaller fleets, and tighter profit margins, which means even small improvements in decision making can lead to significant operational gains. By transforming raw operational data into actionable insights, KeretaSewa AI helps operators make smarter and faster business decisions.

- **Increase Monthly Rental Revenue**
  By identifying pricing opportunities, high demand periods, and underutilized vehicles, operators can improve earnings from existing fleet assets without necessarily increasing fleet size.
- **Reduce Idle Vehicle Time**
  Vehicles that remain unused generate depreciation and maintenance costs without producing income. Better allocation and promotional strategies can reduce unnecessary idle days.
- **Improve Fleet Utilization**
  Existing vehicles can be used more efficiently through optimized pricing, targeted promotions, and better demand matching, maximizing return on investment.
- **Support Faster and More Confident Decisions**
  Instead of relying solely on intuition, operators can use AI-backed recommendations with clear reasoning and expected impact to make decisions more confidently.
- **Improve Operational Visibility**
  Centralized dashboards and analytics allow owners to quickly understand performance trends, weak areas, and growth opportunities.
- **Encourage Data Driven Business Culture**
  The platform promotes a shift from reactive decision making toward proactive planning based on measurable insights.

Strategic Fit / Impact

KeretaSewa AI strongly aligns with the target domain of AI for Economic Empowerment and Decision Intelligence by showing how artificial intelligence can directly improve SME business performance in a practical real world setting.

The project delivers strategic impact through the following areas:

- **Interpretation of Structured Business Data**
  The system processes rental records, pricing data, utilization metrics, and booking trends into understandable performance insights.
- **Context Aware Reasoning Using AI**
  Recommendations are generated based on the operator's own fleet conditions, demand patterns, and business performance rather than generic advice.
- **Recommendation of Actions and Strategies**
  The platform suggests practical next steps such as pricing adjustments, promotions, or reallocation of underperforming vehicles.
- **Explainable Decision Outputs**
  Each recommendation includes reasoning, confidence level, trade offs, and estimated impact, allowing users to understand why the suggestion was made.

- **Measurable Economic Impact**
  The system focuses on tangible outcomes such as higher revenue, lower idle time, better fleet utilization, and stronger profitability.
- **Scalable Cross Industry Potential**
  While initially designed for car rental operators, the same concept can be extended to logistics fleets, equipment rentals, tourism transport, and other asset based SMEs.

3. Product Purpose

3.1 Main Goal of the System

To provide a simple, affordable, and intelligent decision support platform that helps SME car rental operators maximize fleet utilization and improve revenue performance using AI-powered recommendations.

3.2 Intended Users (Target Audience)

- Small to medium car rental operators (5–50 vehicles)
- Independent rental business owners
- Multi branch local rental businesses
- Operators using manual or semi digital workflows
- Businesses without dedicated analytics teams

4. System Functionalities

4.1 Description

KeretaSewa AI is an internal decision support platform for rental operators. It analyzes uploaded operational data and provides recommendations to improve pricing, fleet utilization, and revenue performance.

4.2 Key Functionalities

i. Dashboard Analytics

The dashboard provides a centralized and visual overview of current business performance, allowing operators to quickly understand how their fleet is performing.

Main analytics include:

- **Fleet Utilization Overview**
  Displays how efficiently each vehicle is being used over a selected period.

- **Revenue Trends**
  Shows current and historical revenue performance for easier monitoring.
- **Booking Frequency Trends**
  Tracks how often vehicles are rented and identifies demand patterns.
- **Weekday vs Weekend Demand Analysis**
  Highlights rental differences between weekdays and weekends to support pricing and promotion strategies.
- **Idle Vehicle Detection**
  Identifies vehicles with long idle periods that may require repositioning, discounts, or promotional campaigns.
- **Performance Visibility**
  Allows operators to compare top performing and underperforming vehicles at a glance.

ii. AI Recommendation Engine

The AI Recommendation Engine analyzes fleet data and automatically generates practical business suggestions to improve performance.

Example recommendations include:

- Increase weekend pricing for high-demand vehicles
- Promote underutilized vehicles through targeted offers
- Reallocate vehicles between branches or locations
- Maintain pricing for already strong-performing assets
- Bundle long-term rental packages for slower-moving vehicles

Each recommendation is accompanied by:

- **Reasoning** : Why the recommendation is suggested
- **Expected Impact** : Potential increase in revenue, bookings, or utilization
- **Confidence Score** :Estimated confidence level of the suggestion
- **Trade-off Analysis** :Possible downside or risk of implementation
- **Priority Level** :Which actions should be considered first

This helps users not only receive suggestions, but also understand how to act on them confidently.

iii. Ask AI Assistant

KeretaSewa AI includes an interactive AI assistant that allows operators to ask business-related questions in natural language.

Users may ask:

- “Should I accept this booking?”
- “What if I reduce weekday pricing?”
- “Which vehicle should I promote next?”
- “Why is my utilization low this month?”
- “How can I increase monthly revenue?”

The AI assistant responds with:

- Clear answers
- Supporting reasoning
- Suggested actions
- Decision guidance based on uploaded fleet data

This reduces the need for manual analysis.

iv. Data Input System

The system supports flexible methods for entering operational data so that SMEs with different levels of digital readiness can use the platform easily.

Supported methods include:

- **CSV Upload**
  Users can upload spreadsheets containing booking records, vehicle data, pricing, or rental history.
- **Manual Data Entry**
  Users may enter vehicle and booking information directly through forms.
- **Future Integration Potential**
  Can later be extended to connect with booking systems or third party platforms.

v. Report Generation

The platform allows users to generate downloadable PDF reports based on the current dashboard view.

Included in the report:

- Fleet performance summary
- Revenue and utilization metrics
- Charts and analytics visuals
- AI recommendations
- Projected business impact

This is useful for internal review, management meetings, and business planning discussions.

vi. Account & Profile Management

The platform includes basic account management features to support secure access and personalized usage.

Features include:

- **User Registration**
  New users can create an account using email and password.
- **User Authentication**
  Registered users can securely sign in to access their dashboard.
- **Profile Management**
  Users may update display name and account details.
- **Password Management**
  Users can change their password securely.
- **Account Deletion**
  Users may permanently delete their account after confirmation.
- **Session Logout**
  Users can sign out safely from the platform.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

The platform uses OpenRouter API with Qwen3-32B as the core language model due to its strengths in reasoning, summarization, recommendation generation, and explainable responses.

4.3.2 Prompting Strategy

KeretaSewa AI uses a structured and task-specific prompting strategy to ensure that the AI produces responses that are relevant, consistent, and useful for business decision-making.

Instead of asking the model open ended questions, the system provides the model with a clear business role: it acts as a fleet strategy assistant for car rental operators. The prompt guides the model to focus only on the provided fleet data, such as revenue, utilization rate, idle days, booking count, vehicle performance, and weekday versus weekend demand. This reduces the chance of irrelevant or hallucinated responses.

The prompting strategy is divided into three main decision-support purposes:

- **Recommendation generation**
  The AI analyzes fleet performance and produces practical business suggestions, such as improving utilization, adjusting pricing, reducing idle days, or promoting underperforming vehicles.
- **Conversational decision support**
  Users can ask business related questions in natural language, and the AI responds with a clear verdict, reasoning, recommended actions, and key takeaway.

To keep the output consistent, the prompts also define the expected response structure, including action, reasoning, estimated impact, confidence level, and trade-off. This makes the AI's recommendations easier for users to understand and compare.

4.3.3 Context & Input Handling

- **Structured CSV Data Prioritized**
  The system prioritizes structured CSV uploads containing booking records, vehicle details, rental prices, and transaction history, as these formats are easier to validate and analyze.
- **Essential Metrics Extracted Before Inference**
  Before sending data to the AI model, the system derives key business indicators such as fleet utilization rate, booking frequency, revenue totals, idle vehicle days, and weekday versus weekend demand patterns.
- **Focused Business Context Provided to AI**
  Only relevant summarized metrics are passed into the model prompt to improve response quality and reduce unnecessary noise.
- **Scalable Data Handling (Future Enhancement)**
  For larger datasets, future versions may summarize or aggregate records before inference to maintain efficiency and lower processing cost.
- **Consistency Across Inputs**
  Whether data is uploaded through CSV or entered manually, the system standardizes inputs into a common internal format for analysis.

4.3.4 Fallback & Failure Behavior

AI Returns Invalid JSON / Malformed Output:

- Display a clear dashboard error message instead of broken content
- Prevent frontend crashes or blank screens
- Allow the user to manually retry using the refresh function
- Preserve the previous successful recommendation set when available
- Log the failure for debugging and future improvement

If Ask AI Assistant Fails:

- Return a user friendly error message
- Suggest retrying after a short moment
- Keep the chat interface responsive without freezing

If API Service is Temporarily Unavailable:

- Continue displaying dashboard analytics that do not depend on AI
- Show the last successful recommendations with timestamp indicators
- Inform the user that intelligent features are temporarily unavailable

If Uploaded Data is Invalid:

- Prompt the user to correct missing or unsupported fields

5. User Stories & Use Cases

User Stories

- As a rental operator, I want to know which vehicles are underutilized so I can improve fleet usage and reduce idle time.
- As a business owner, I want pricing recommendations so I can maximize revenue during high demand and low demand periods.
- As an operator, I want to ask business questions in natural language so I can make faster decisions without manual analysis.
- As a user, I want clear reasoning behind recommendations so I can trust the AI suggestions.

Main Use Cases

UC 1: Upload Fleet Data

Actor	Rental Operator
Description	The user uploads operational data such as vehicle models, booking history, pricing, and utilization records through CSV upload or manual input.
Pre-condition	The rental operator has accessed the KeretaSewa AI platform and possesses basic fleet or booking data for analysis.
Post-condition	The uploaded data is stored in the system, processed successfully, and ready for dashboard analytics and AI recommendation generation.
Main flow	1. The system displays the data upload page. 2. The rental operator selects either CSV upload or manual data entry. 3. The rental operator uploads the CSV file or enters fleet data manually. 4. The system validates required fields such as vehicle model, pricing, and utilization data. 5. The system processes the uploaded data and generates summary metrics. 6. The rental operator is redirected to the dashboard page.
Alternative Flow	If manual input is selected, the system allows the user to add records one by one.
Exception Flow	1. If the uploaded file format is invalid, the system displays an error message and requests a supported CSV file. 2. If required fields are missing, the system prompts the user to correct the data before proceeding.

UC 2: View Dashboard Analytics

Actor	Rental Operator
Description	The user views business performance metrics and trends through the dashboard.
Pre-condition	Fleet data has been uploaded and processed successfully.
Post-condition	The user gains visibility into operational performance and identifies areas for improvement.
Main flow	1. The user opens the dashboard page. 2. The system displays utilization rate, revenue, bookings, and idle days. 3. The system shows weekday versus weekend demand trends. 4. The user reviews underperforming vehicles.
Alternative Flow	-
Exception Flow	1. If no data exists, the system prompts the user to upload data first. 2. If dashboard loading fails, an error message is displayed.

UC 3:Ask AI Assistant

Actor	Rental Operator
Description	The user asks natural-language business questions to receive decision support.
Pre-condition	Fleet data has been uploaded and processed successfully.

Post-condition	The user receives an AI response containing verdict, reasoning, and suggested actions.
Main flow	1. The user opens the Ask AI assistant. 2. The user enters a business question. 3. The system sends the question together with fleet context to OpenRouter API using Qwen3-32B. 4. The AI analyzes the request. 5. The system displays a structured response.
Alternative Flow	-
Exception Flow	1. If the question is unrelated, the system redirects the user to fleet topics. 2. If AI response fails, an error message is shown.

UC 4: Refresh AI Recommendations

Actor	Rental Operator
Description	The user requests the system to regenerate updated recommendations based on the current fleet data.
Pre-condition	Fleet data is already loaded in the dashboard.
Post-condition	Updated recommendations are displayed to the user.
Main flow	1. The user clicks the refresh recommendations button. 2. The system resends current fleet data to the recommendation model through OpenRouter API. 3. The AI generates updated recommendations. 4. The system replaces the previous recommendations with the new results.

Alternative Flow	-
Exception Flow	If AI service is unavailable, an error message is displayed.

UC 5: Add New Fleet Data

Actor	Rental Operator
Description	The user adds new vehicle or booking data to update the dashboard and AI insights.
Pre-condition	The user is on the KeretaSewa AI dashboard.
Post-condition	New data is added, dashboard metrics are updated, and recommendations can be refreshed.
Main flow	1. The user clicks “Add Data”. 2. The system redirects the user to the data input page. 3. The user enters or uploads new fleet data. 4. The system updates the processed fleet data. 5. The user returns to the dashboard. 6. Updated metrics are displayed.
Alternative Flow	The user cancels data entry and returns to the dashboard without changes.
Exception Flow	If the input data is incomplete, the system requests correction.

UC 6:Generate Report

Actor	Rental Operator
Description	The user generates a PDF report based on the current dashboard view, including fleet performance metrics, charts,

	AI recommendations, and projected impact summary.
Pre-condition	Fleet data has been uploaded and dashboard analytics are displayed.
Post-condition	A PDF version of the dashboard report is generated and downloaded for the user.
Main flow	1. The user clicks the “Generate Report” button. 2. The system captures the current dashboard view. 3. The system converts the dashboard content into a PDF file. 4. The system downloads the generated report to the user’s device.
Alternative Flow	-
Exception Flow	If the dashboard content cannot be captured, the system displays an error message.

6. Features Included (Scope Definition)

The current prototype of KeretaSewa AI focuses on delivering core decision support capabilities for SME car rental operators. Included features are:

- **Dashboard Analytics**
  Visual dashboard showing revenue, bookings, utilization rate, idle days, and vehicle performance trends.
- **AI Recommendation Engine**
  Automatically generates practical business recommendations based on uploaded fleet data.
- **Ask AI Assistant**
  Interactive AI assistant that answers business-related questions in natural language.

- **CSV / Manual Data Upload**
  Supports both spreadsheet upload and manual data entry for flexible onboarding.
- **Revenue & Utilization Estimation**
  Provides projected revenue improvement and fleet efficiency impact based on current recommendations.
- **Explainable Recommendations**
  Each recommendation includes reasoning, expected impact, confidence level, and trade-off analysis.
- **PDF Report Generation**
  Allows users to generate downloadable PDF reports based on the current dashboard view.
- **Refresh Recommendation Function**
  Users may regenerate updated AI recommendations using the latest available data.
- **Account Management**
  Supports user registration, secure sign in, profile updates, password changes, account deletion, and sign out functionality.

7. Features Not Included (Scope Control)

To maintain realistic prototype scope and development focus, the following features are not included in the current version:

- **Full Booking Website Replacement**
  The system does not function as a customer facing reservation platform.
- **Payment Gateway Integration**
  Online payment collection and invoicing are not included.
- **Live GPS Vehicle Tracking**
  Real time location tracking and telematics integration are outside the current scope.
- **Multi Branch Enterprise Management**
  Advanced enterprise features such as centralized branch coordination are not implemented.
- **Automated Operational Execution**
  The platform provides recommendations only and does not automatically apply pricing or allocation changes.

8. Assumptions & Constraints

Assumptions

The development of KeretaSewa AI is based on several practical assumptions regarding user readiness, data availability, and business operations. These assumptions help define the scope of the current prototype and expected system behavior.

1. Basic Fleet Data Availability

It is assumed that rental operators have access to basic operational data such as:

- Vehicle model names
- Base rental prices
- Booking dates
- Rental prices per booking
- Booking completion status

These represent the minimum data required for generating dashboard insights and AI recommendations.

2. Data Accuracy

The system assumes that uploaded data is reasonably accurate, updated, and relevant. Incorrect, duplicated, outdated, or incomplete records may reduce the reliability of recommendations and analytics outputs.

3. User Willingness to Adopt AI Support

It is assumed that operators are open to using AI generated recommendations as decision support tools, while still applying their own business judgment before implementation.

Constraints

The current prototype is subject to several practical and technical limitations.

1. Prototype Scope Constraint

KeretaSewa AI is currently focused on decision intelligence and internal business analytics. It does not include:

- Full customer booking website functionality
- Online payment gateway integration
- Customer relationship management (CRM) modules
- Full fleet operations management system

2. Data Dependency Constraint

Recommendation quality depends heavily on the quality, quantity, and consistency of uploaded business data. Poor data quality may lead to weaker insights.

3. Internet / API Dependency Constraint

AI powered features such as recommendations and Ask AI depend on internet connectivity and model service availability. Temporary outages may affect intelligent features.

9. Risks & Questions Throughout Development

1. User Trust and Adoption Risk

Some SME operators may hesitate to trust AI generated business advice, especially if they are accustomed to relying on personal experience or intuition.

Mitigation: Recommendations are supported by reasoning, confidence levels, expected impact, and trade-off explanations to improve transparency and user confidence.

2. API Availability Risk

The system depends on OpenRouter API service availability for recommendation generation and Ask AI responses. Temporary outages, rate limits, or latency may interrupt AI-powered features.

Mitigation: Display the most recent successful recommendations with timestamps, provide clear user facing error messages, and consider multi model providers for future production deployment.

3. Data Quality Risk

Poor quality uploaded data, missing values, or outdated booking records may reduce recommendation accuracy and dashboard reliability.

Mitigation: The current prototype assumes basic valid data input. Future versions may include validation checks, missing value alerts, and automated data cleaning support.

4. Scalability Risk

As user numbers or datasets grow, AI response time and processing cost may increase.

Mitigation: Future versions may optimize prompts, summarize large datasets, and introduce tiered processing strategies.

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